COMMUNICATION SKILLS II

MEIC & METI

WEEK 1

HOW MISUNDERSTANDINGS HAPPEN: PERCEPTION, ASSUMPTIONS, MENTAL MODELS

Andresa Oliveira

TODAY’S CLASS

- Presentation of Teacher and Teacher Assistants
- Presentation and Structure of the Course
- Some Rules
- Topic 1: How Misunderstandings Happen: Perception, Assumptions, Mental Models

COMMUNICATION SKILLS II

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COMMUNICATION SKILLS II

COURSE DESCRIPTION

The course is **strongly practice-based**:

- Each session combines ≤30 minutes of **theory.**
- With **structured active practice** – students will practice communication as happens in real life and professional contexts – through structured exercises, simulations, and collaborative tasks.
- Students progressively build a personal "**Communication Toolkit**" through weekly **artifacts** that can be reused in academic, professional, and everyday contexts.

COMMUNICATION SKILLS II

COURSE DESCRIPTION

The course is designed to be **inclusive** for students with introverted profiles and/or social anxiety, using “write-first” strategies, gradual exposure, and predictable participation formats.

COMMUNICATION SKILLS II

COURSE DESCRIPTION

This course develops **advanced communication competencies** for future engineers by focusing on the **psychological mechanisms** behind effective interaction.

Week	Topic
Week 1 (25, 26, 27 de February)	How Misunderstandings H Assumptions, Mer
Week 2 (4, 5, 6 March)	Emotion in Communic Regulation, De-E
Week 3 (11, 12, 13 March)	Psychological Safety & T Communication Dynamic Work
Week 4 (18, 19, 20 March)	Stakeholder Com
Week 5 (25,26,27 March)	Communication fo
	Influence Without

COMMUNICATION SKILLS II

ASSESSMENT METHODS

There will be **no written test or exam** for this course. Instead, you will be assessed through **three in-class evaluation moments**, scheduled only between **Week 2 and Week 5**. Be sure to save the dates and plan accordingly. All assessments are **practical and aligned with what we train in class**.

Important: All evaluation moments include an **oral component**. Students are required to deliver short oral presentations/simulations in class as part of Assignment 1, Assignment 2, and Assignment 3.

Participation requirement: In the group assessments (Assignment 2 and Assignment 3), **every student must speak** in class. Each student is required to deliver **at least one oral contribution per assignment** (e.g., presenting one section).

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ASSESSMENT METHODS

To pass the course, you must achieve:

- A **final grade of at least 9.5/20** (Final Grade (0–20) = 0.30 × Assignment 1 + 0.30 × Assignment 2 + 0.40 × Assignment 3), and
- Participation in all evaluation moments (or an officially accepted justification).

All assignments are assessed **in class** between **Week 2 and Week 5** (inclusive).
Only Assignment 3 involves **preparation outside the classroom** (partially).

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ASSESSMENT METHODS

ASSIGNMENT 1 | COMMUNICATION DEBUGGING

Contributes 30% to final grade.

Setup: individual; written, 20 minutes; oral presentation, max 1 min.

Note: Students receive a short scenario/message in class.

You must:
- Identify assumptions / possible misunderstandings (mental models, ladder of inference).
- Rewrite the message for clarity and psychological safety.
- Propose 2–3 clarification / de-escalation lines.
- Deliver a short oral explanation (maximum duration: 1 minute) summarizing: what was misunderstood, what was changed, and why the new version is safer.

Dates: Week 2

COMMUNICATION SKILLS II

ASSESSMENT METHODS

ASSIGNMENT 2 | MEETING TO DECISION

Contributes 30% to final grade.

Setup: 5-6 student groups; oral presentation (meeting simulation 5 min + decision log and team norms 5 min).

Note: Students receive a short scenario in class.

You must:
- Run a short, structured meeting simulation (5 min).
- Present a Decision Log and 5 Team Communication Norms (5 min).

All group members must speak during the meeting simulation, at least once!

Dates: Week 4

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ASSESSMENT METHODS

ASSIGNMENT 3 | STAKEHOLDER COMMUNICATION SIMULATION

Contributes 40% to final grade.

Setup: 5-6 student groups; oral presentation.

Note: student groups receive a scenario in advance (Week 4).

Outside class, you must prepare:
- A 1-page Stakeholder Update (progress, impact, risks, uncertainty, next steps).
- A 90-second oral script with the message you will deliver (no slides, but you can bring notes, it's not a memory test).
- Speaking roles (mandatory). Plan who says what, so everyone speaks.

In class, you:
- Do an oral Stakeholder Update (90-second), based on the script prepared outside class.
- Receive a "stakeholder question" (given on spot) and answer orally (1 min).
- Receive a "complaint" (given on spot) and do an oral repair (1 min).

All group members must speak during the Stakeholder Update, at least once!

Dates: Week 5.

SOME RULES

This class is a safe place!

- **Respectful communication:** disagreement is welcome, disrespect is not.
- No student is required to share sensitive personal experiences.
- **Feedback is specific and behavioral** (what worked / what to improve), never humiliating.
- **Practice is part of learning:** mistakes are treated as data for improvement.

SOME RULES

Channels of Communication

FÉNIX = grades, class slides, announcements
EMAIL = individual questions

Doubts and Questions

Please let us know!

TODAY’S CLASS

Theory:

- Communication is cultural
- Mental models
- Common ground
- Attribution errors
- Ladder of inference
- Business Communication: communicating “so work happens”
- Message channels

Active Practice:

- Misunderstanding Lab
- Ladder of Inference Drill

Artifact:

- Clarity Checklist

AKNOWLEDGMENTS

Elaborated also with materials from lectures by previous Teachers: Sofia Sá and Claudia Silva.

START QUESTIONS

If two engineers read the same Jira ticket, why do they often build two different things?

When someone is late replying, do you assume ‘they don’t care’ or ‘they’re overloaded’? Why is your brain so confident?

What creates more misunderstandings: too little information… or too many hidden assumptions?

TODAY'S CLAIM

WE RESPOND TO OUR INTERPRETATION OF REALITY, NOT REALITY ITSELF

- Misunderstandings rarely come from "lack of intelligence".
- They come from how humans build meaning:
  select → interpret → conclude → act
- Engineers aren't immune (they're just faster at reasoning from partial data)

JUST TO REMEMBER

COMMUNICATION

Includes:

- Talking.
- Listening.
- Asking questions.
- Answering questions.
- Non-verbal aspects.
- Silence (yes, it does communicate).

All we do, inside and outside the “engineering world”, implies communication.

ADVANCE NOTICE

COMMUNICATION IS CULTURAL

Will you marry me?

Silence

- In **English**, it would be interpreted as **uncertainty**.
- In **Japanese**, it would be interpreted as **acceptance**.
- In **Igbo** (Southeastern Nigeria), it would be considered as **denial** if the woman were to continue to stand there, and **acceptance** if she ran away.

"Communication between cultures" by Samovar, Porter and McDaniel (2009)

MENTAL MODELS

WHY “SAME WORDS” ≠ “SAME MEANING”

Mental Models

internal representations of how something works (people, systems, tasks)

- They guide what we notice and what we ignore.
- Two people can look at the same message and “see” different problems.

MENTAL MODELS

WHY “SAME WORDS” ≠ “SAME MEANING”

“OK”

- Version 1: The person is busy and agrees.
- Version 2: The person is upset and disengaging.
- Version 3: The person is confused but doesn’t want conflict.

Common ground matters

- Communication becomes easier when people share mutual knowledge/beliefs/assumptions and know they share it.

COMMOM GROUND

ISN’T AUTOMATIC: IT’S BUILT

- Clark & Brennan describe **conversation as collaborative**, requiring “grounding”:
  - We constantly update a shared understanding (“Do we both mean the same thing?”).
  - Grounding is reached when participants mutually believe understanding is “good enough to move on”.

Implication:

If you don’t explicitly build common ground, people fill gaps with assumptions.

THE ATTRIBUTION TRAP

WHY WE MISREAD PEOPLE'S INTENTIONS

- A common **bias in social perception**: we over-explain others' behavior with personality ("they're careless") and underweight context ("they had constraints").

- Classic evidence:
  - Jones & Harris (1967): people inferred an author's "true attitude" even when the author had limited choice about what to write.
  - Ross (1977): describes this tendency as central to attribution errors ("intuitive psychologist" distortions).

In teamwork:

This bias turns delays/mistakes into character judgments instead of process questions.

LADDER OF INFERENCE

WHY WE CLIMB FROM DATA TO CERTAINTY (SO FAST)

- A useful model for **metacognition**:
  - We select data → add meaning → make assumptions → form conclusions → adopt beliefs → act

- It explains why two people can act differently from the same situation.

Key point:
The ladder isn't "bad"—it's unavoidable.
The skill is noticing when you climbed too fast and checking the lower rungs.

“DEBUGGING” HABIT

SLOW DOWN THE LADDER

- First and foremost:
  - Know yourself (what do you think and believe in).
  - Acknowledge that your look at the world is not universal.
  - Find out as much as you can about people you communicate with (and what lies behind their assumptions and beliefs).

“DEBUGGING” HABIT

SLOW DOWN THE LADDER

- Practical questions:
  - What did I observe vs. what did I infer? (facts vs story)
  - What assumption am I making about intent?
  - What information would change my conclusion?(testability)
  - What’s the simplest clarification question I can ask? (grounding)

STOP MISUNDERSTANDINGS

3 MICRO-SCRIPTS

01

Clarify meaning:
“When you say X, do you mean A or B?”

02

Check assumptions:
“I may be assuming ___ — is that correct?”

03

Ground next step:
“So decision is ___. Owner is ___. Deadline is ___.”

BUSINESS COMMUNICATION

COMMUNICATE SO WORK HAPPENS

- In professional contexts, communication isn’t self-expression — it’s coordination:
  - clarity of goal
  - alignment on decisions
  - ownership and next steps

Goal:

You’re making sure understanding is “good enough to move forward.”

MESSAGE CHANNELS

CHOOSING THE RIGHT CHANNEL REDUCES MISUNDERSTANDINGS

01

Media Richness Theory (Daft & Lengel)

- Media differ in “richness” (feedback speed, cues, personalization)
- Richer media tend to help with equivocal/ambiguous issues; leaner media suit straightforward information.

02

Media Synchronicity Theory (Dennis et al., MISQ)

- Different media support different processes:
  - *Conveyance* (sharing information) often works well with lower synchronicity media (asynchronous)
  - *Convergence* (reaching shared meaning/decision) benefits from higher synchronicity media (real-time).

MESSAGE CHANNELS

PRACTICAL TRIAGE RULE

Use this decision rule:

- If the task is ambiguous / conflict-prone / needs alignment → choose **high-feedback channel** (quick meeting, call, live discussion).
- If the task is stable / informational / needs traceability → choose **written async** (ticket, doc, email).
- Then **document the decision** in writing (so work happens).

ACTIVE PRACTICE

ACTIVITY 1

MISUNDERSTANDING LAB

Goal: show how interpretations diverge + practice grounding and clarity.

Setup: groups of 5–6.

Steps:

1. Misunderstanding Lab (15 min)
2. Clarity Rewrite (20 min)
3. Fast Gallery Walk (10 min)

YOU’VE GOT A TICKET

“Can you fix the latency issue ASAP? It’s getting worse.”

You must produce:

- 3 different interpretations (what “latency issue” means, what “fix” means, what “ASAP” means).
- Assumptions behind each interpretation.

CLARIY REWRITE

“Can you fix the latency issue ASAP? It’s getting worse.”

You must rewrite it into an implementable message that includes:

- context, definition, scope, impact.
- explicit request + owner + deadline.
- choose channel (ticket/email/meeting/chat) + one sentence why (ambiguity vs need for convergence).

GALLERY WALK

“Can you fix the latency issue ASAP? It’s getting worse.”

Post your rewrites in a slide (wall).

Half the teams stay as hosts; half circulate.

Circulators leave two comments on two other groups using:

- “What I understand the task is…”
- “What’s still ambiguous/assumed…”
- “One question that would build common ground…”

ACTIVITY 2

LADDER OF INFERENCE DRILL

Goal: train the “debug your thinking” habit

Setup: first solo and then pairs.

Steps:

1. Ladder template (5 min, solo)
2. Partner challenges (5 min, pairs)
3. Clarify-before react line (15 min)

YOU’VE GOT A CONFLICT

A teammate replies “OK” to your detailed plan and then does something different.

You must fill a ladder template:

- data I noticed → meaning I made → assumptions → conclusion → action I want to take.

YOU’VE GOT A CONFLICT

A teammate replies “OK” to your detailed plan and then does something different.

Find a partner and challenge your partner (choose 2-3 questions):

- “What else could this mean?”
- “What data are you missing?”
- What evidence do you have for yourconclusion? What evidence do you not have?
- What context could explain this without bad intent?
- What information would change my mind?
- “What’s a neutral clarification question?”

YOU’VE GOT A CONFLICT

A teammate replies “OK” to your detailed plan and then does something different.

Each pair choose and share with class:

- 1 line to clarify-before-react.

ARTIFACT

CLARITY CHECKLIST

USE BEFORE REPLYING, ESCALATING, OR MAKING A DECISION

CLARITY CHECKLIST

01

Separate facts from interpretations

- What did I observe (exact words/data) vs. what did I infer?
- What might I be missing?

02

Surface hidden assumptions

- What does each key term mean (e.g., “fix”, “urgent”)?
- What am I assuming about scope, priority, constraints, dependencies?
- What am I assuming about the other person’s intent?

CLARITY CHECKLIST

03

Build common ground with clarifying questions

Use questions that reduce ambiguity and mind-reading:

- “When you say X, do you mean A or B?”
- “I might be assuming ___- is that correct?”
- “What is the impact and what metric/example defines the issue?”
- “What does ASAP mean (hours/days)?”
- “What is in scope / out of scope?”
- “What are the constraints/dependencies?”
- “What would ‘good enough to proceed’ look like?”

CLARITY CHECKLIST

04

Close the loop: confirm alignment so work happens

- “So the goal is ___.”
- “Decision: ___.”
- “Owner: ___.”
- “Deadline: ___.”
- “Next checkpoint: ___ (time + channel).”
- “I’ll summarize this in ___ (ticket/email/doc).”

CLARITY CHECKLIST

05

Choose the right channel (message triage)

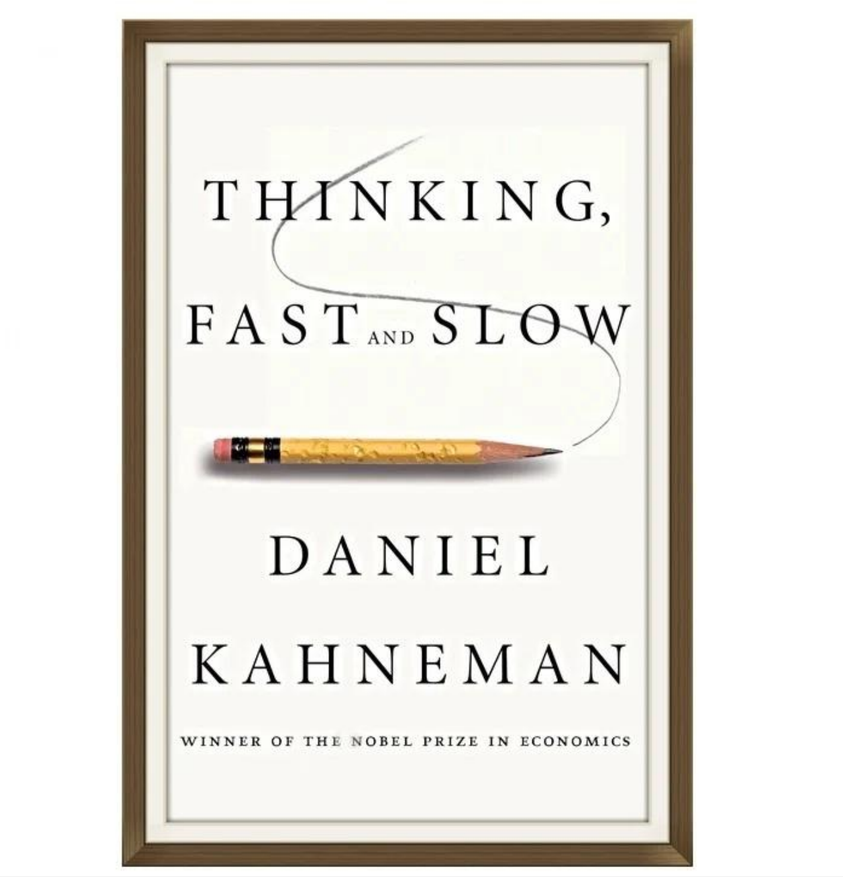
- **Meeting/call** when you need alignment, convergence, or fast clarification (high ambiguity/sensitivity).
- **Ticket/doc/email** when you need traceability, detail, and asynchronous processing (clear tasks/info).
- **Chat** for quick coordination, but move to richer channels if ambiguity grows.
- After synchronous discussion, document decisions in writing.

READING SUGGESTIONS

READING SUGGESTION

It explains why your brain confidently climbs the ladder of inference: we make fast interpretations (System 1), then we rationalize them (System 2). It's a good explanation of snap judgments, overconfidence, and biased interpretation—all major drivers of misunderstandings.

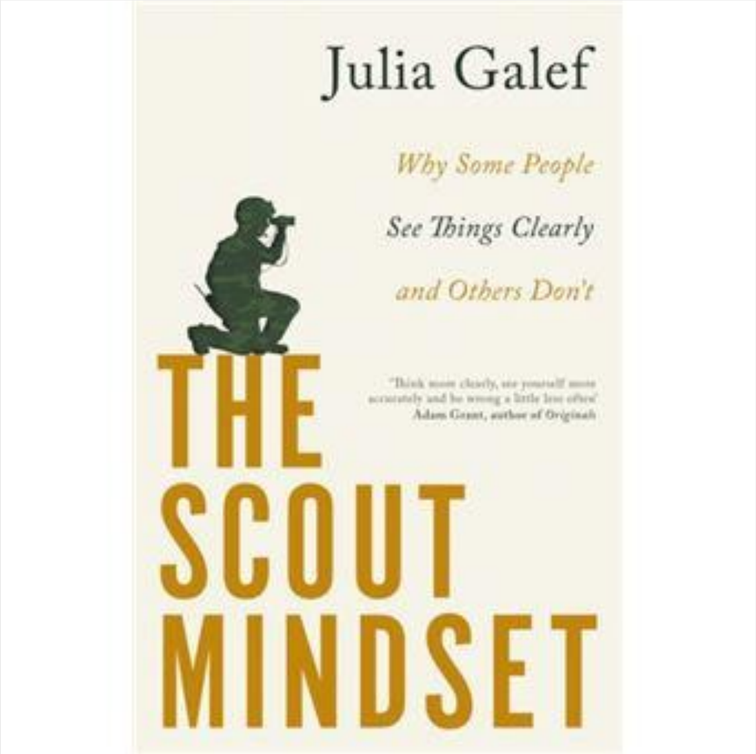
Best parts for this week class: the early chapters on System 1 vs System 2, and chapters on heuristics/biases (how we fill gaps with stories).



READING SUGGESTION

It’s a practical guide to the class claim: **we react to our interpretation, not reality.** Galef contrasts “soldier mindset” (defending a conclusion) with “scout mindset” (curious, truth-seeking), which maps directly onto assumption-checking and building common ground instead of mind-reading.

Best parts for this week class: the opening chapters that define scout vs soldier mindset + the concrete tactics for testing assumptions and staying curious under uncertainty.



THANK YOU